

David Lindsey

590 W Highway 105 #128 • Monument, CO 80132 | 850-879-4105 | davidlindsey5@gmail.com

Director, Assembly, Integration & Test (AI&T)

Augustus Aerospace — Lone Tree, CO | September 2023 – January 2026

Led assembly, integration, verification, testing, and on-orbit checkout of advanced space payloads through the full mission lifecycle, contributing to the successful launch, commissioning, and operational deployment of flight systems.

Developed and configured complex LEO ISR mission simulation environments to enable realistic, high-fidelity validation of payload performance; reduced pre-launch risk, verified mission-specific functionality, and ensured operational readiness under representative mission conditions.

- Directed payload integration, functional verification, and environmental qualification of RF and digital systems to ensure mechanical, electrical, and software compatibility with flight platforms.
- Led a multidisciplinary team of six engineers in the development and delivery of complex spaceflight payloads supporting a \$10M multi-year government contract.
- Designed and implemented mission-level GNSS simulation environments modeling orbital dynamics, signal propagation, and timing. Integrated hardware-in-the-loop testing using Skydel GNSS simulation software, NovAtel OEM719 receivers, and Ettus USRP radios to validate mission functionality. Resolved complex signal-generation challenges by adjusting transmitted parameters to prevent jamming and spoofing detection on the NovAtel receiver, ensuring stable and accurate simulation performance.
- Utilized COSMOS Command Control and Communication software suite to strengthen test fidelity, reduce on-orbit risk, and streamline mission verification through automation and repeatable, realistic scenarios.
- Integrated all required ground support equipment into COSMOS to enable automated payload testing and improve test repeatability.
- Collaborated with Department of Energy laboratory partner to test and troubleshoot ground station hardware, software, and mission planning systems, ensuring reliable end-to-end functionality and interoperability across mission operations.
- Led preparation and successful execution of Ground Readiness Review (GRR) and Day-In-The-Life (DITL) demonstrations, validating end-to-end payload, network, and ground system readiness under mission-representative conditions.
- Developed and executed payload-level test plans encompassing thermal, vibration, EMI/EMC, and post-environmental validation to ensure full mission readiness and compliance with operational standards.
- Utilized a comprehensive simulation environment integrating ground station, RF link, spacecraft/payload EDU, and GNSS/SOI simulators to conduct complete mission testing using the same tools employed for flight operations. This approach enabled early identification and resolution of latent system issues that would have otherwise gone undetected, directly contributing to mission success.
- Led comprehensive pre-launch test campaigns, coordinating mission-like operational demonstrations for customer and government review to verify system performance and mission readiness.
- Led RF testing of field unit receivers, verifying link budget calculations and conducting simulated contact tests to demonstrate the system's ability to reliably acquire and track mission signals.

- Coordinated with design, production, and platform teams to maintain accurate ICDs, integration procedures, and payload documentation.
- Worked with the mission operations team to identify opportunities to reduce operational risk and enhance performance, reliability, and usability in future payload designs.
- Executed relocation and redesign of the integration and test laboratory, ensuring operational continuity and enhanced capability.
- Designed and laid out a new electronics laboratory, creating a modular, reconfigurable workspace optimized for diverse payload configurations and evolving program requirements.
- Managed the design and construction of a new \$1.2 million headquarters.

Project Manager

Architectural Arts & MFI — Des Moines, IA | 2020 – 2023

- Directed commercial architectural millwork projects from contract award through installation, managing scope, schedule, budget, and client expectations.
- Managed a portfolio of up to 12 concurrent projects with individual contract values ranging from approximately \$10K to \$200K.
- Maintained profit margins above company averages through proactive planning, cost control, and effective risk management.
- Identified and resolved design, procurement, and production issues early, minimizing schedule impacts and reducing costly rework.
- Coordinated clients, architects, vendors, and manufacturing teams to deliver high-quality projects on schedule.

Software Support Technician / IT & Design Engineer

Hexagon Manufacturing Intelligence & Architectural Arts | 2017 – 2020

- Supported engineering software systems; managed ERP data integration; designed CNC programs for custom manufacturing.
- Troubleshoot SQL Server, network connectivity, and CAD/CAM issues to restore operations and improve reliability.

Earlier Experience (Condensed)

Woodlane Cabinet Co. & Florida State University | 2002 – 2017

Progressive technical and engineering roles involving software integration, design automation, and lab research. Conducted chemistry research at Florida State University, leading to a publication in the Journal of Materials Chemistry C (2016), building foundational problem-solving and interdisciplinary research skills.

Education

Tallahassee Community College — A.A., Pre-Chemical Engineering, 2010–2013

Technical Skills

- RF & SDR: RF communications, SDR testing, antenna evaluation, GNSS simulation, NovAtel OEM719 receivers, Ettus USRP radios, RF connector and waveguide familiarity (SMA, N-type, BNC), spectrum analyzer and signal generator operation, benchtop RF testing, RF data collection, and test-procedure execution.
- Test & Evaluation: Use of electronic test and measurement equipment to support laboratory operations; evaluation of COTS components; documentation/reporting of test results and hardware anomalies; mechanical, electrical, and RF troubleshooting.
- Platforms: Windows, Linux, VMware, COSMOS, MATLAB, PBX, Web Servers, Docker
- Programming: Python, MATLAB, Ruby
- Networking: Fiber-optic infrastructure, network configuration, secure/air-gapped operations, VPN, and site-to-site connectivity
- Software Tools: Microsoft Office Suite, Skydel, ERP (INNERGY), CAD/CNC, Wireshark
- Laboratory & Operations: Assembly, configuration, test, and repair of hardware per documentation; method development and process control improvements; adherence to ESD and safety standards; maintaining clean, organized work areas; supporting facility operations.

Core Competencies

- Technical Problem Solving
- Cross-functional Leadership
- Systems Integration
- Assembly, Integration & Test (AI&T)
- Risk Management
- Program Execution
- Customer & Stakeholder Communication
- Requirements Management
- Root Cause Analysis
- Process Improvement
- Technical Documentation
- Schedule & Cost Management
- Team Development

Research, Publications & Awards

- Co-author: Modulating the Electrical Conductivity of Metal Organic Framework Films with Intercalated Guest π -Systems, Journal of Materials Chemistry C, 2016
- NSF S-STEM Scholar • NSF REU Participant • CRC Press Chemistry Achievement Award • Phi Theta Kappa Honor Society